Yeon-Ji Song

+82-10-5179-4255 | yjasz98@gmail.com | yeonjisong.github.io

Education

Seoul National University (SNU) 2021.09 – 2027.09 (expected)
PhD in Interdisciplinary Program in Neuroscience & AI Institute *Seoul, South Korea*
- Advisor: Prof. Byoung-Tak Zhang

Hong Kong University of Science and Technology (HKUST) 2017.09 – 2021.06
BEng in Electronic and Computer Engineering *Clear Water Bay, Hong Kong*

Research Interests

Visual generation via physical concept grounding, including moving objects and camera dynamics, in a neurosymbolic way.
Dynamic scene understanding for novel view synthesis and reconstruction from blurry monocular inputs.
Object-centric learning in the physical world with time-static object appearance and time-varying object motion.
Robotics and embodied AI leveraging learned object and scene representations for robotics tasks in real-world settings.

Publications

*equal contribution, †corresponding author

Kinematics-Driven Gaussian Shape Deformation for Blurry Monocular Dynamic Scenes
Yeon-Ji Song, Kiyoung Kwon, Junoh Lee, Jin-Hwa Kim†, Byoung-Tak Zhang†
in Proceedings of ICML 2026

Relaxed Rigidity with Ray-based Grouping for Dynamic Gaussian Splatting
Junoh Lee, Junmyeong Lee, Yeon-Ji Song, Inhwan Bae, Jisu Shin, Hae-Gon Jeon†, Jin-Hwa Kim†
in Proceedings of ECCV 2026

One-shot 3D Affordance Learning for Multi-stage Robotic Manipulation
Hyunseo Kim, Yeon-Ji Song, Minsu Lee†, Byoung-Tak Zhang†
in IEEE Access 2026

OCK: Unsupervised Dynamic Video Prediction with Object-Centric Kinematics
Yeon-Ji Song, Jaein Kim*, Suhyung Choi*, Jin-Hwa Kim†, Byoung-Tak Zhang†
in Proceedings of ICCV 2025

DBMovi-GS: Dynamic View Synthesis from Blurry Monocular Video via Sparse-Controlled Gaussian Splatting
Yeon-Ji Song, Jaein Kim, Byoungju Kim, Byoung-Tak Zhang†
in Proceedings of CVPR 2025 Workshop on Neural Fields Beyond Conventional Cameras

Continuous SO(3) Equivariant Convolution for 3D Point Cloud Analysis
Jaein Kim, Heebin Yoo, Dong-Sig Han, Yeon-Ji Song, Byoung-Tak Zhang†
in Proceedings of ECCV 2024

Unsupervised Visual Dynamics Learning with Multi-Object Kinematics
Yeon-Ji Song, Byoung-Tak Zhang†
in Proceedings of KCC 2024

Learning Object Appearance and Motion Dynamics with Object-Centric Representations
Yeon-Ji Song, Hyunseo Kim, Suhyung Choi, Jin-Hwa Kim†, Byoung-Tak Zhang†
in Proceedings of NeurIPS 2023 Workshop on Causal Representation Learning

On Discovery of Local Independence over Continuous Variables via Neural Contextual Decomposition
Inwoo Hwang, Yunhyeok Kwak, Yeon-Ji Song, Byoung-Tak Zhang†, Sanghack Lee†
in Proceedings of CLeaR 2023

Professional Service

Program Committee Member (Reviewer)
- NeurIPS 2026, ICML 2026, ICCV 2025, WACV 2024

Technical Mentoring
- AI Youth Challenge (POSCO DX) 2023 – 2026
- Machine Learning and Computer Vision (Hyundai NGV) 2025
- Project XR: AI Chatbot (LognCoding) 2024

Teaching Assistant

Seminars in Neuroscience 1 (SNU)	2026.03 – 2026.06
Seminars in Neuroscience 2 (SNU)	2025.09 – 2025.12
Multimodal Generative AI Theories and Applications (SNU)	2025.09 – 2025.12
Multimodal Deep Learning Theories and Applications (SNU)	2024.09 – 2024.12
Artificial Intelligence (SNU)	2022.03 – 2022.06
New Computer Technology (SNU x HKUST)	2022.03 – 2022.06
ENGG2900D: ROV Community Project (HKUST)	2019.02 – 2022.05

Scholarships and Awards

Samsung Industrial-Academic Scholarship	2025 – 2027
SNU Neuroscience Outstanding Poster Award	2025
KCC 2024 Best Presentation Paper Award	2024
RoboCup@Home DSPL 2nd Place	2022
HKUST Admission Scholarship	2017

Work Experience

Biointelligence Lab (SNU) 2020.12 – 2021.04
Undergraduate Research Intern *Seoul, South Korea*
- Designed and conducted research on Robotics and Reinforcement Learning.

Surromind (SNU) 2020.07 – 2020.09
Artificial Intelligence Research Engineer *Seoul, South Korea*
- Designed and implemented a Deep Learning model for Pose Estimation based on Detectron2.

Robocore AI 2020.06 – 2020.07
Robotics Software Engineer *Science Park, HongKong*
- Created new solutions for temi robots, combining IOT products and the mobility of temi.
- Performed GUI design, system design and solved real-life customer request with AI solution.

Codecrain Inc. 2019.06 – 2019.09
Full-stack Web Developer *Seoul, South Korea*
- Developed a frontend web application along with a senior developer using React.js.
- Implemented React.js and Node.js to enhance functionality and user experience.

Projects

SNU-NAVER Hyperscale AI Center 2023.06 – 2024.05
Student Researcher *SNU*
- Advisor: Jin-Hwa Kim (Leader of Generation Research at NAVER AI Lab)
- Developed video generative models focused on 3D object and motion dynamics.
- Published at ICCV 2025 and NeurIPS 2023 workshop on Causal Representation Learning.

Robot Navigation based on Reinforcement Learning 2020.05 – 2021.05
Final Year Project *HKUST*
- Advisor: Ming Liu (Robotics and Multi-Perception Lab, Robotics Institute)
- Title: Map-based Robot Navigation and Path planning with Deep Reinforcement Learning
- Proposed ML and RL based algorithm for autonomous navigation in a cluttered environment.

Bundleport 2018.05 – 2020.05
CTO & Logistics Manager *HKUST*
- Created a full-stack web application using Node.js, MySQL, HTML5/CSS3, and JavaScript.
- Developed on cloud server using AWS S3, EC2, Elastic Beanstalk and Cloudfront.

HKUST Robotics Team 2018.09 – 2018.12
Robotics Software Engineer *HKUST*
- Designed and implemented algorithms for processing data from Camera and LiDAR sensors.